



Exam Ace Champs

Where Champions Are Made

Quick Revision

Chapterwise

Mind-Maps

BIOLOGY

**class
11**

Covers

all chapters of NCERT

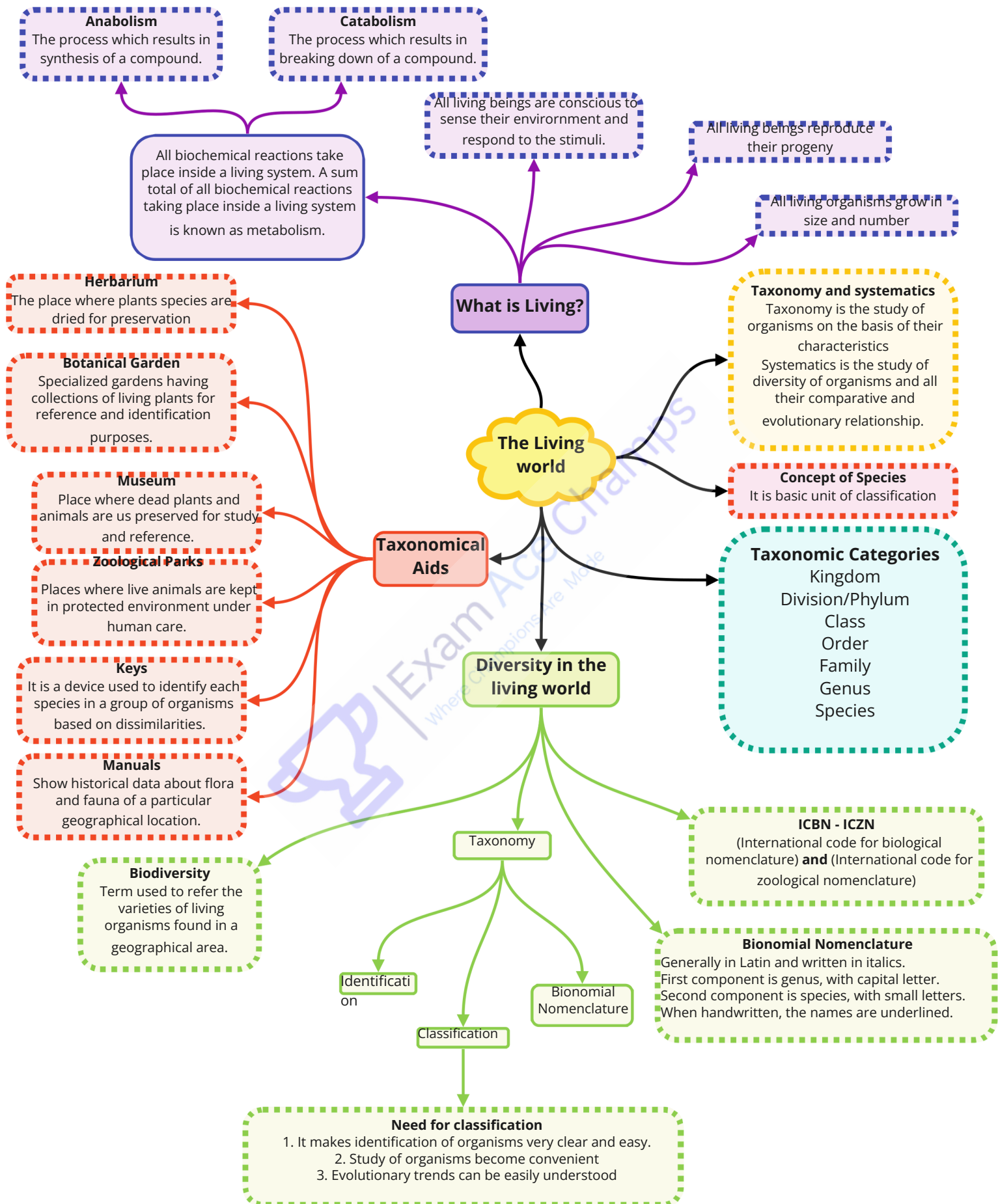


Faster Recall



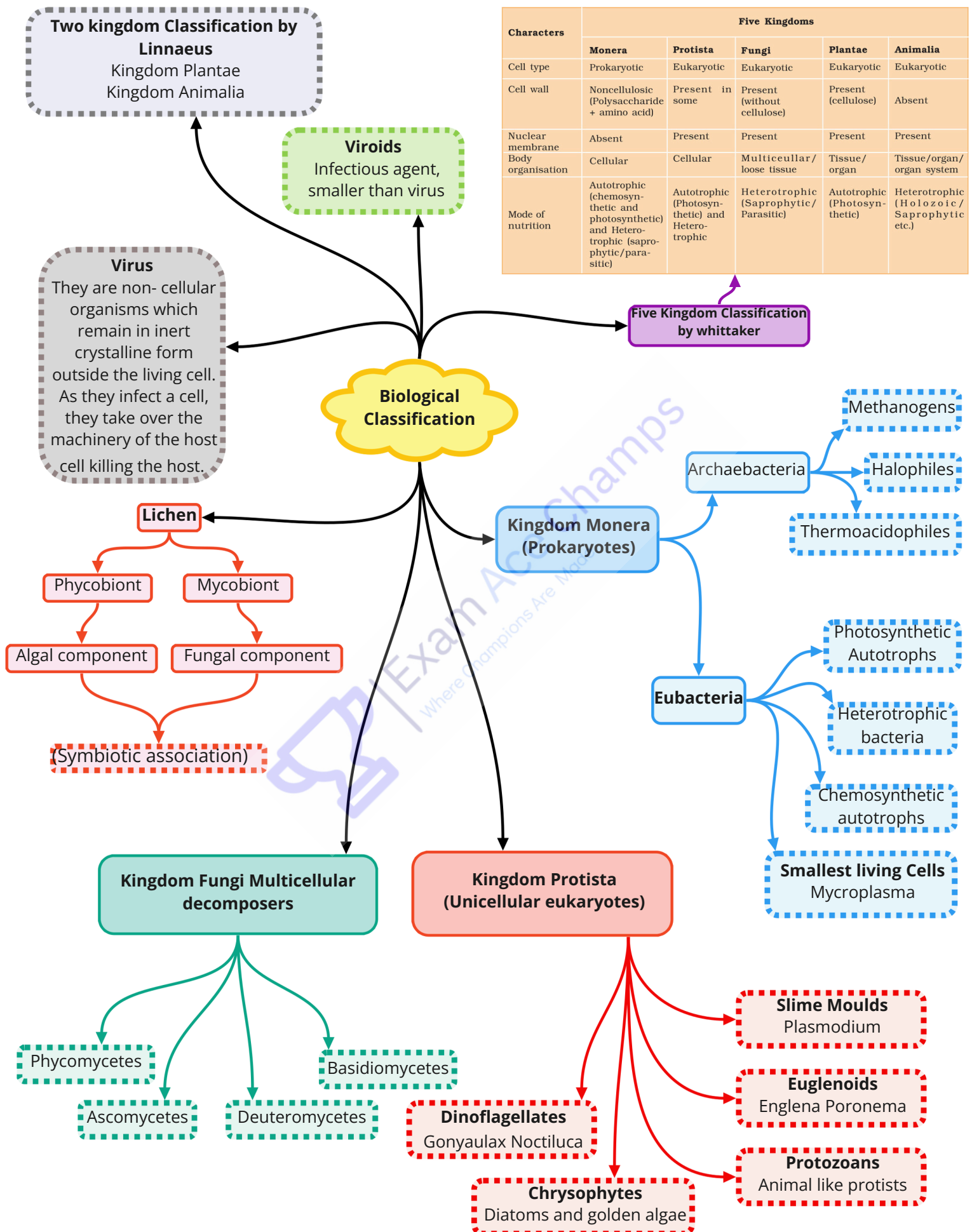
Quick Revision

Class : 11th Biology
Chapter- 1 : The Living world



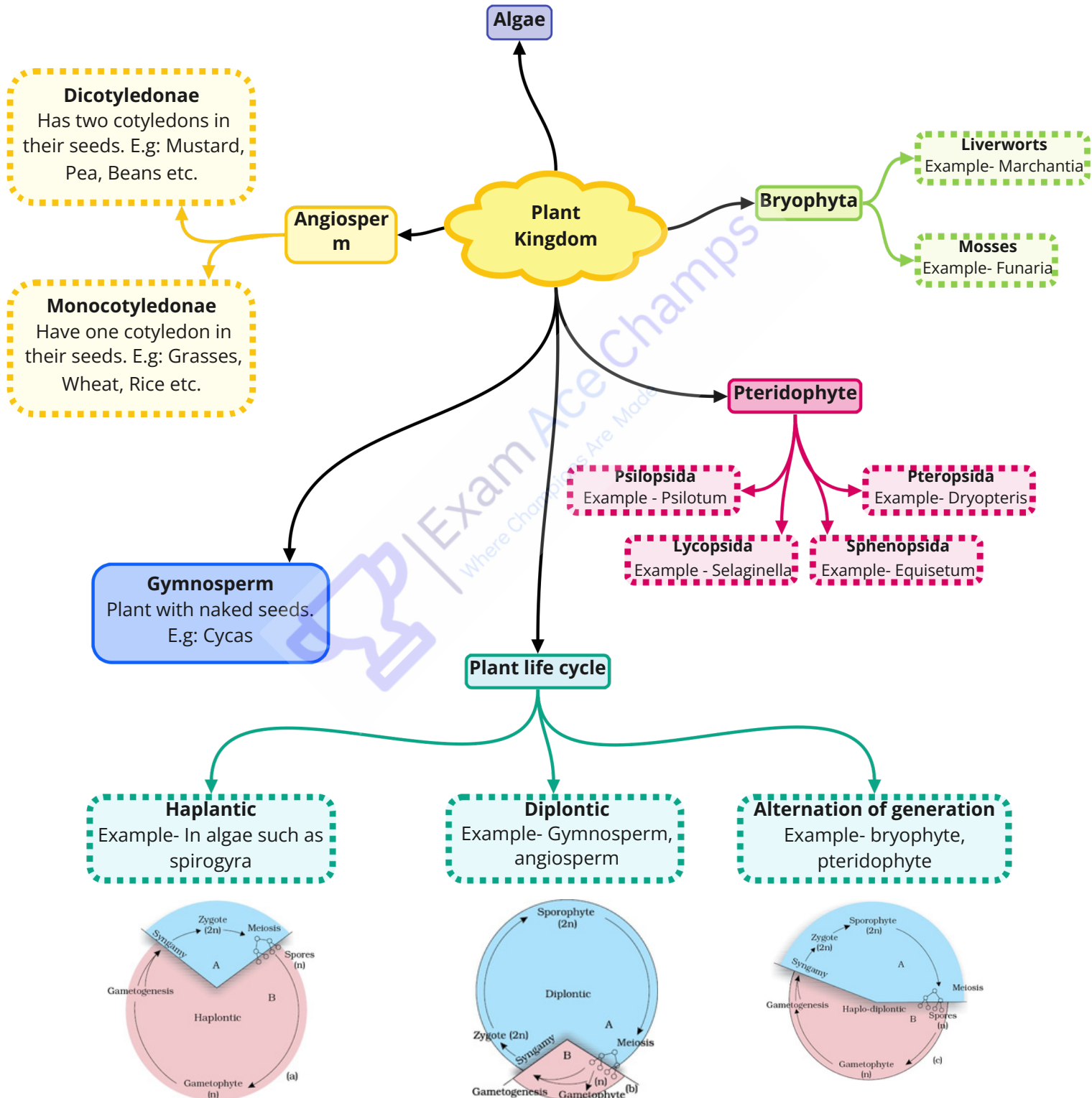
Class : 11th Biology
Chapter- 2 : Biological Classification

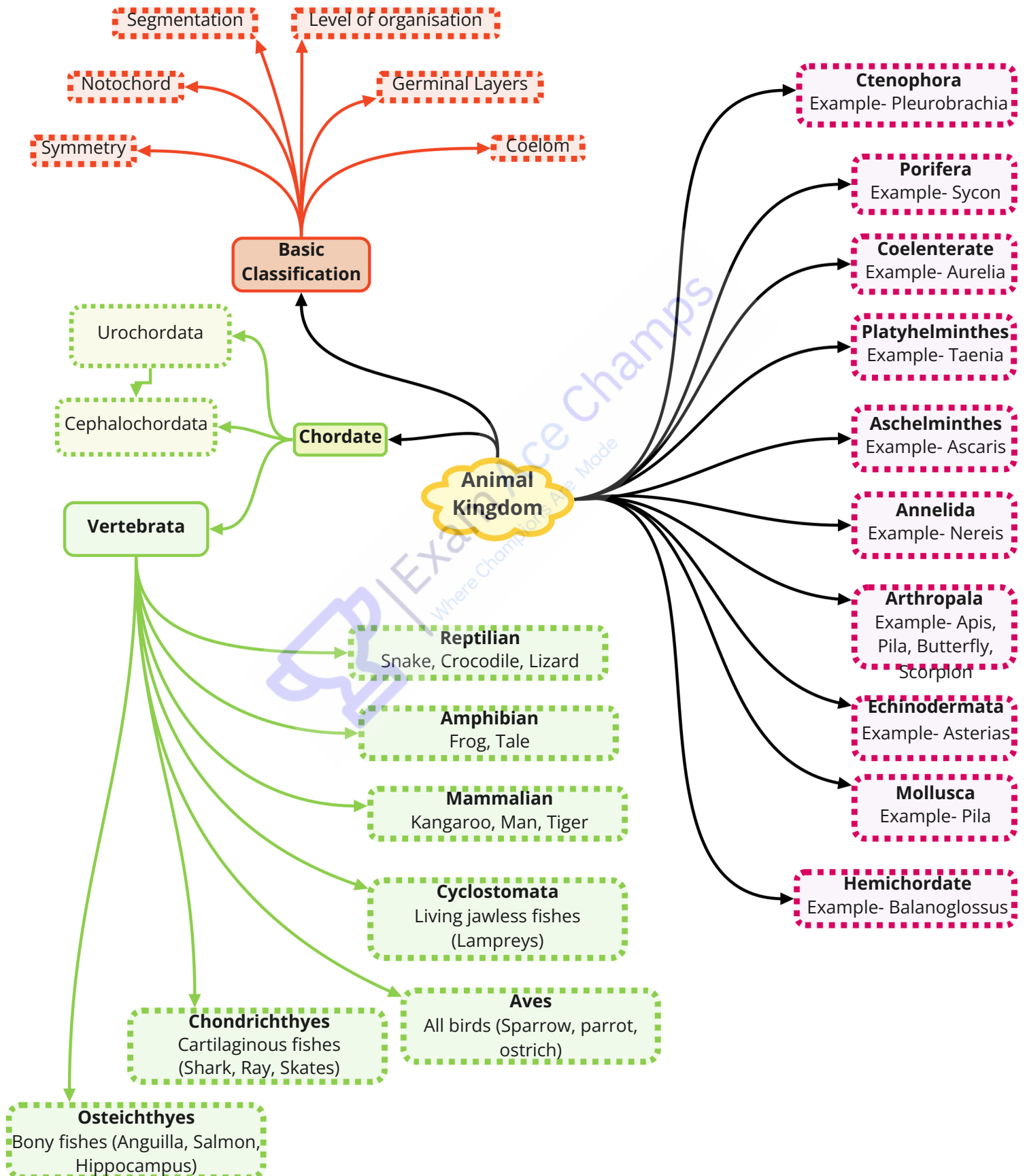
Characters	Five Kingdoms				
	Monera	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Cell wall	Noncellulosic (Polysaccharide + amino acid)	Present in some	Present (without cellulose)	Present (cellulose)	Absent
Nuclear membrane	Absent	Present	Present	Present	Present
Body organisation	Cellular	Cellular	Multicellular / loose tissue	Tissue / organ	Tissue / organ / organ system
Mode of nutrition	Autotrophic (chemosynthetic and photosynthetic) and Heterotrophic (saprophytic/parasitic)	Autotrophic (Photosynthetic) and Heterotrophic	Heterotrophic (Saprophytic / Parasitic)	Autotrophic (Photosynthetic)	Heterotrophic (Holozoic / Saprophytic etc.)



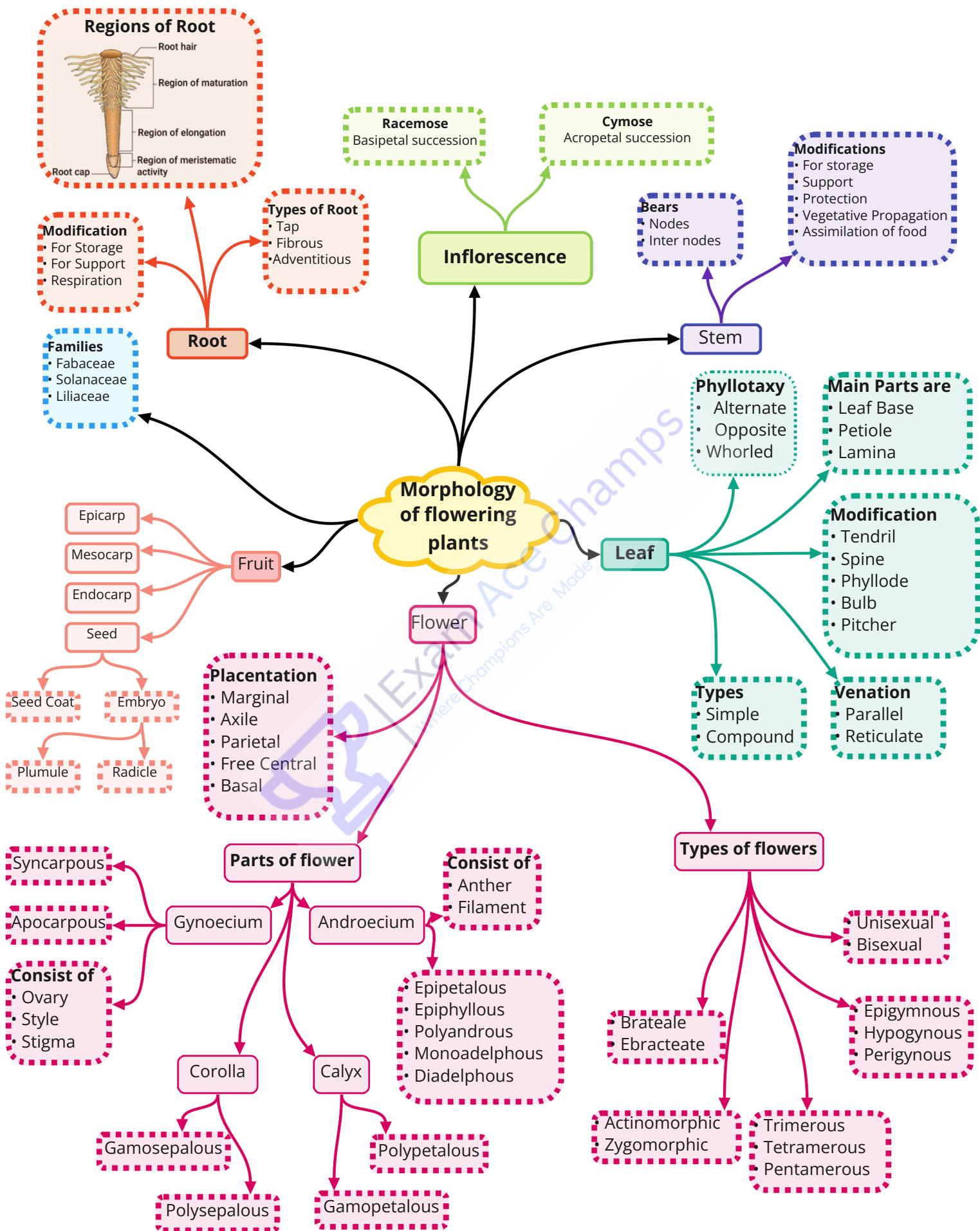
Class : 11th Biology
Chapter- 3 : Plant Kingdom

Classes	Common Name	Major Pigments	Stored Food	Cell Wall	Flagellar Number and Position of Insertions	Habitat
Chlorophyceae	Green algae	Chlorophyll <i>a</i> and <i>b</i> , carotenoids and xanthophyll	Starch and Sugar	Cellulose	2-8, equal, apical	Fresh water, brackish water, salt water
Phaeophyceae	Brown algae	Chlorophyll <i>a</i> and <i>c</i> , fucoxanthin, Flavoxanthin, β carotenes	Mannitol, laminarin	Cellulose and algin	2, unequal, lateral	Fresh water (rare), brackish water, salt water
Rhodophyceae	Red algae	Chlorophyll <i>a</i> and <i>d</i> , phycoerythrin, phycocyanin	Floridean starch	Cellulose	Absent	Fresh water (some), brackish water, salt water (most)

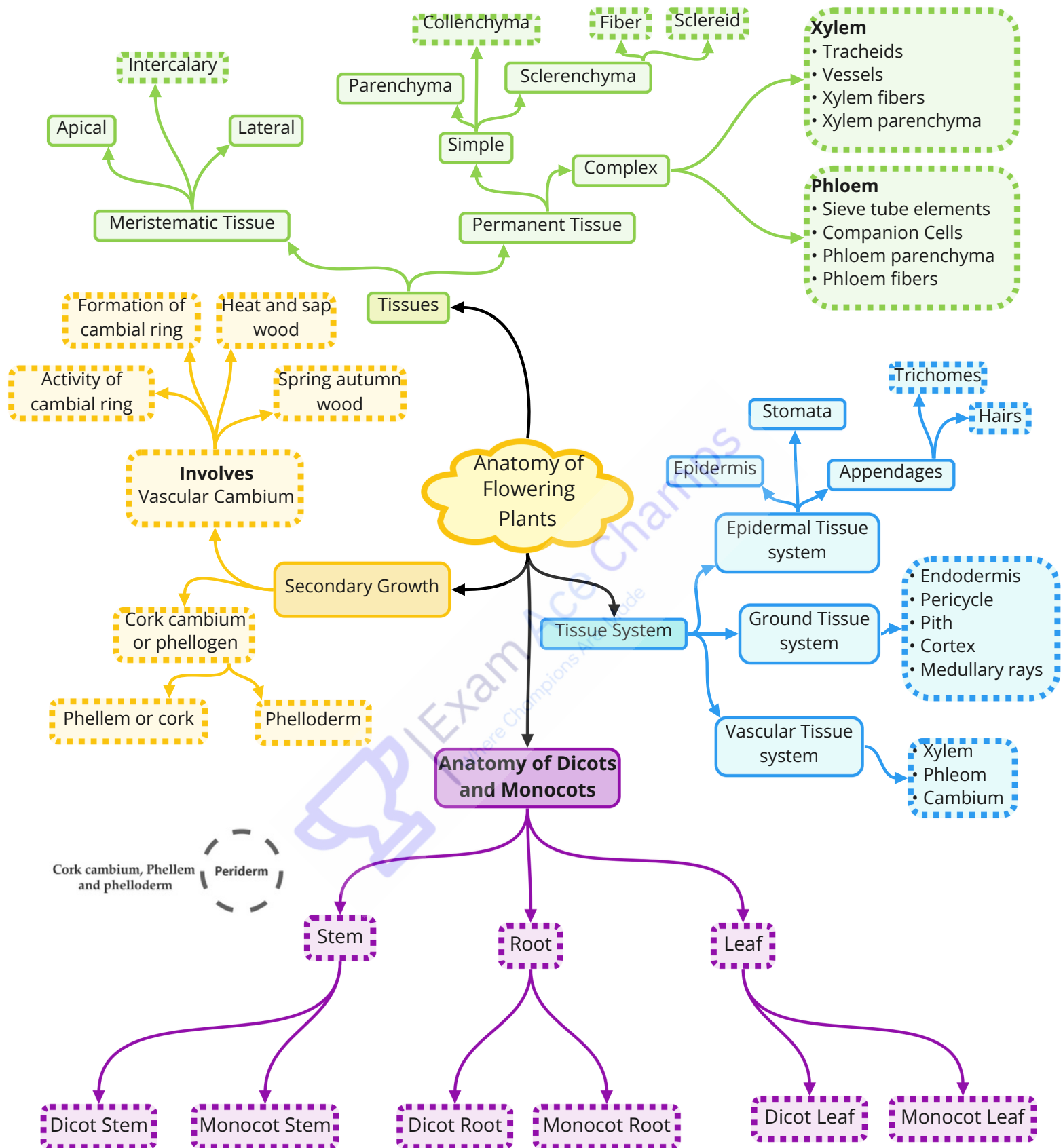




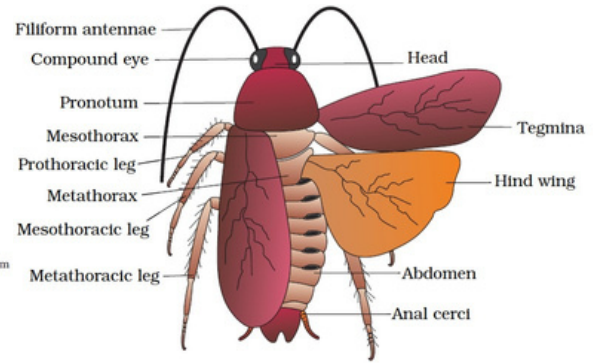
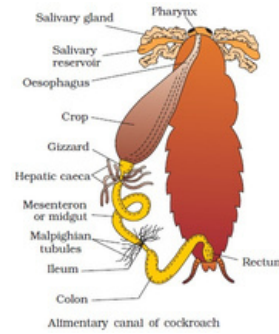
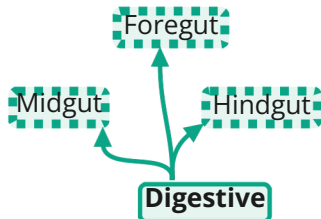
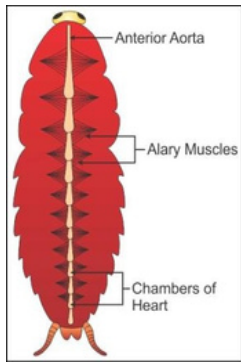
Class : 11th Biology
Chapter- 5 : Morphology Of Flowering Plants



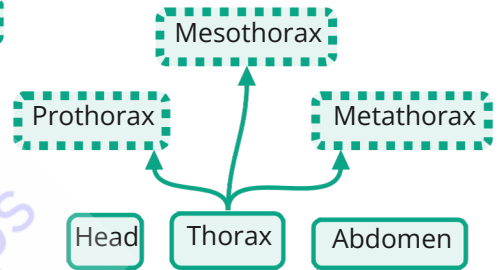
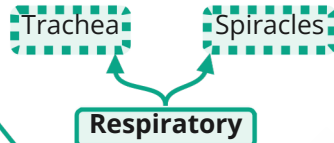
Class : 11th Biology
Chapter- 6 : Anatomy of Flowering Plants



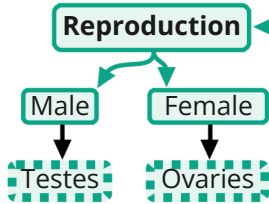
Class : 11th Biology
Chapter- 7 : Structural Organisation in Animals



Circulatory
 Open type- Have poorly developed blood



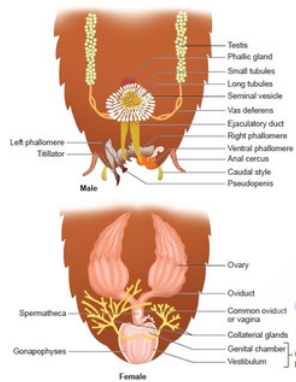
Nervous
 Segmentally arranged ganglia joined by paired longitudinal



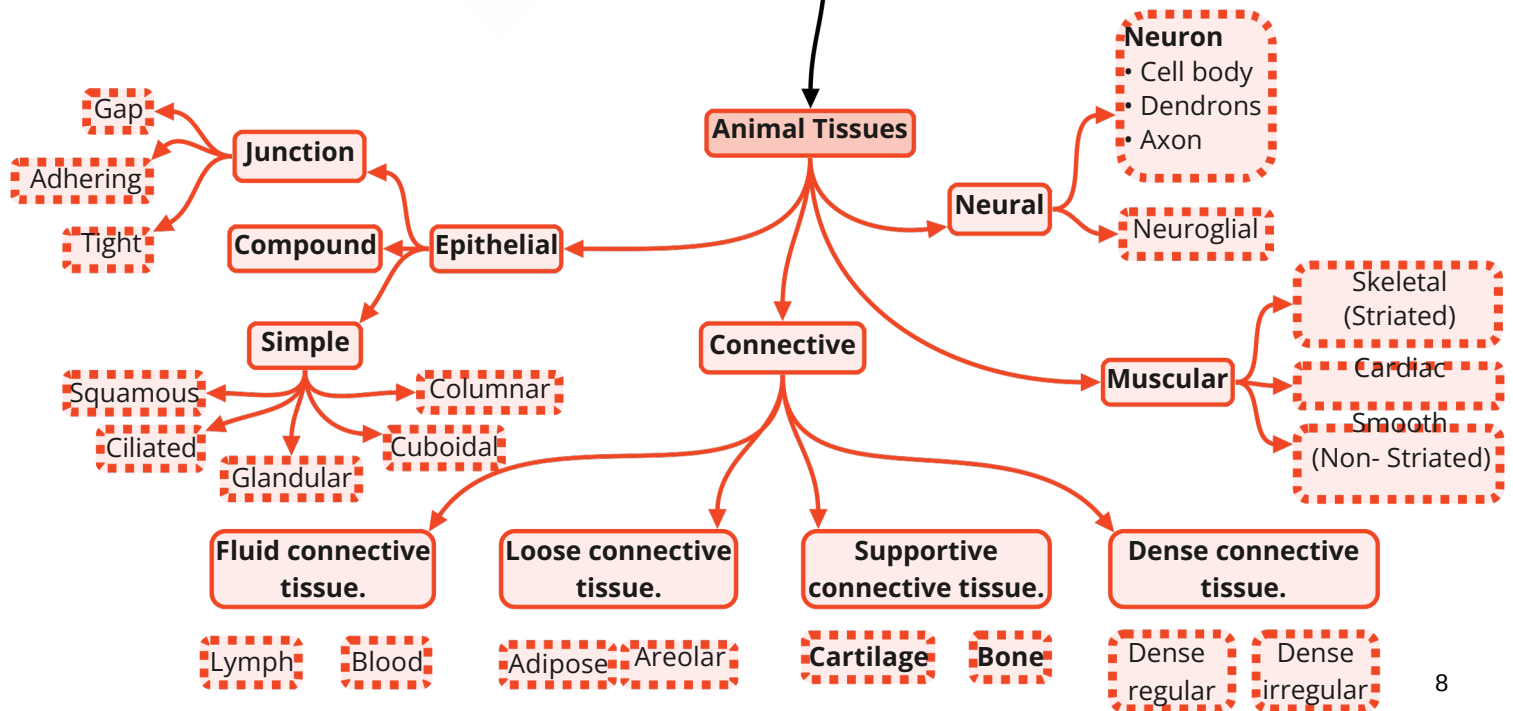
Cockroach (Periplaneta Americana)

Excretion
 Malpighian tubules

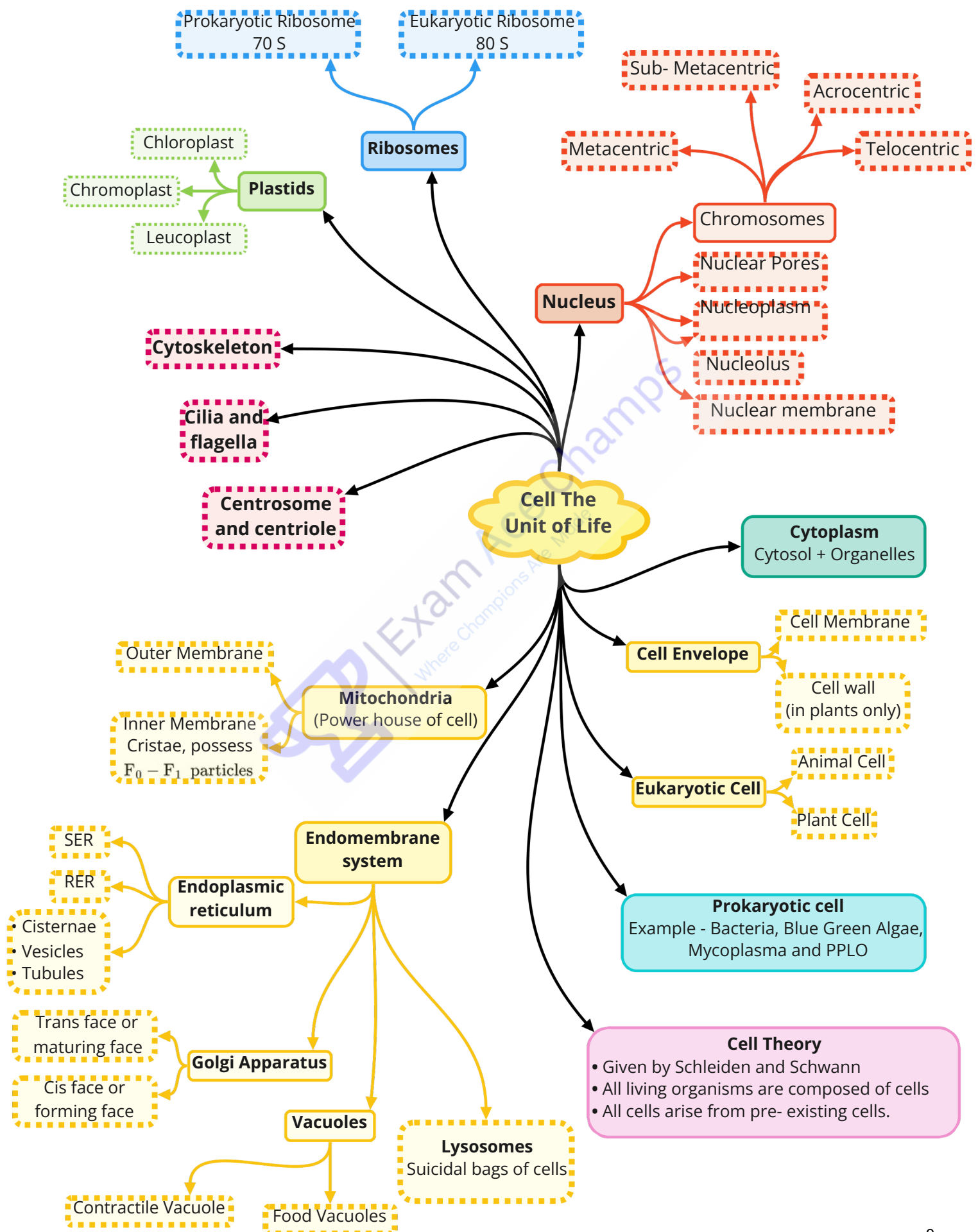
Structural Organisation in Animals

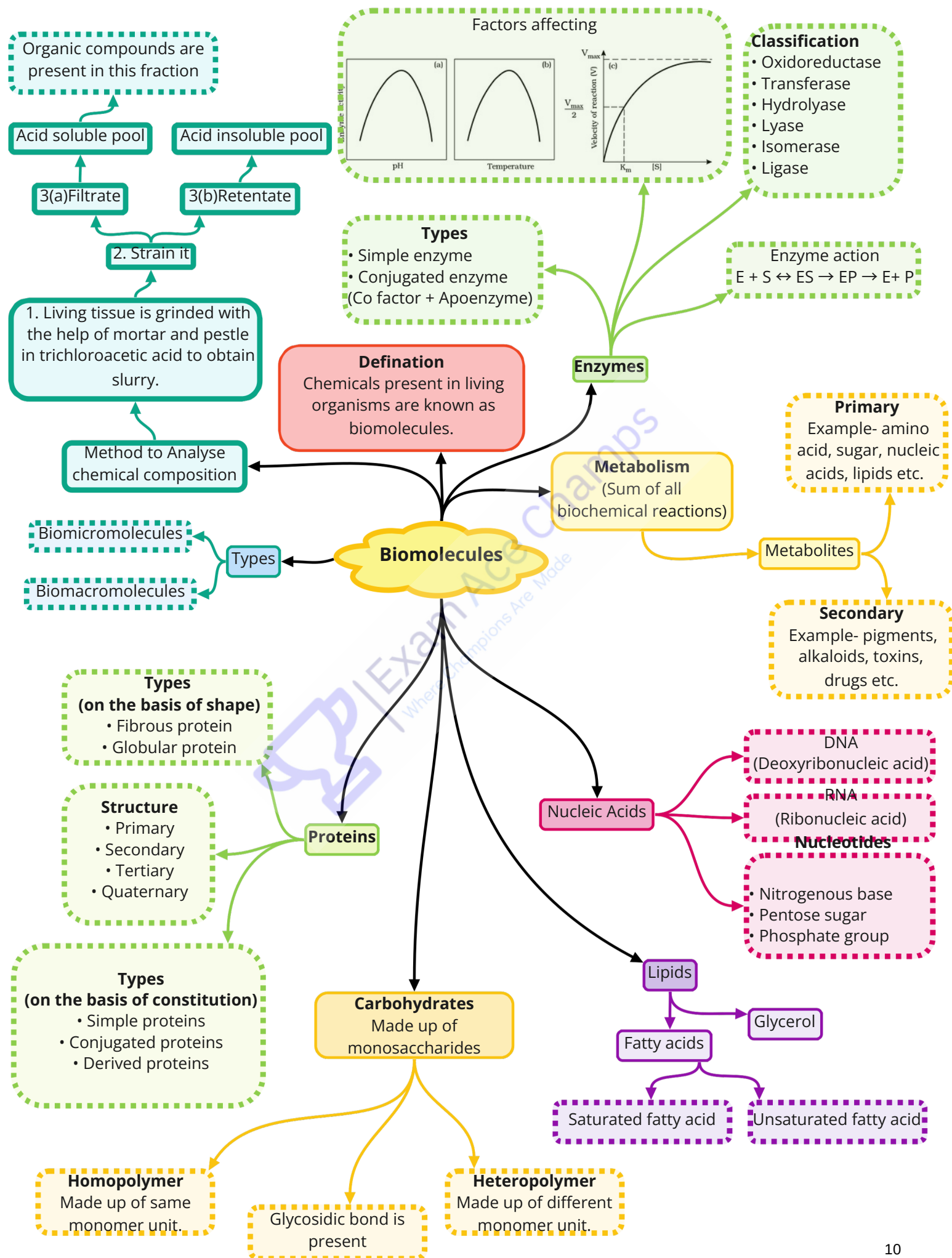


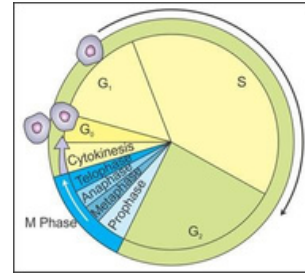
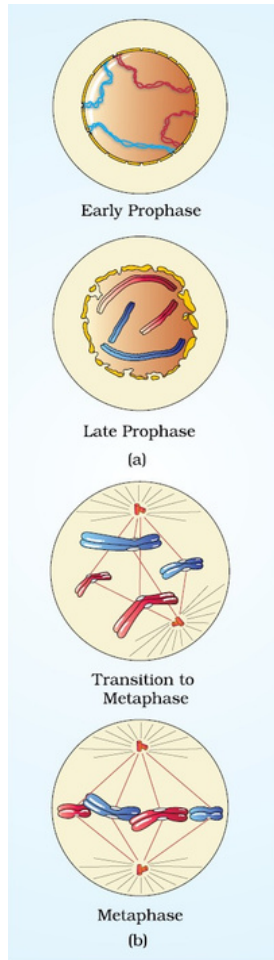
Animal Tissues



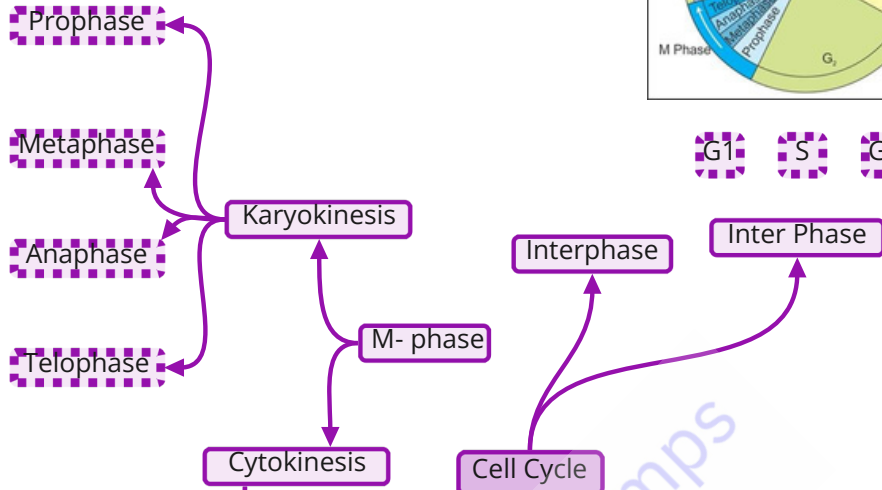
Class : 11th Biology
Chapter- 8 : Cell The Unit of Life



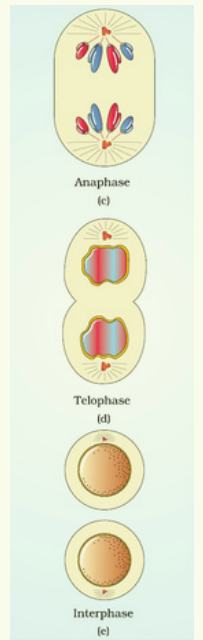




G₁ S G₂



Mitosis- Equational Division

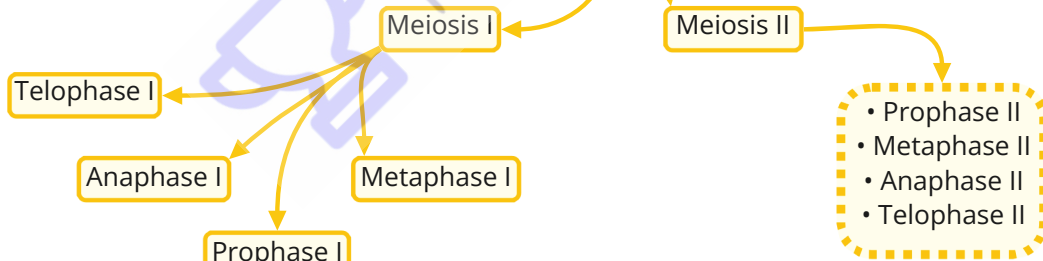


Plant Cell
 Cell wall formation starts in the centre

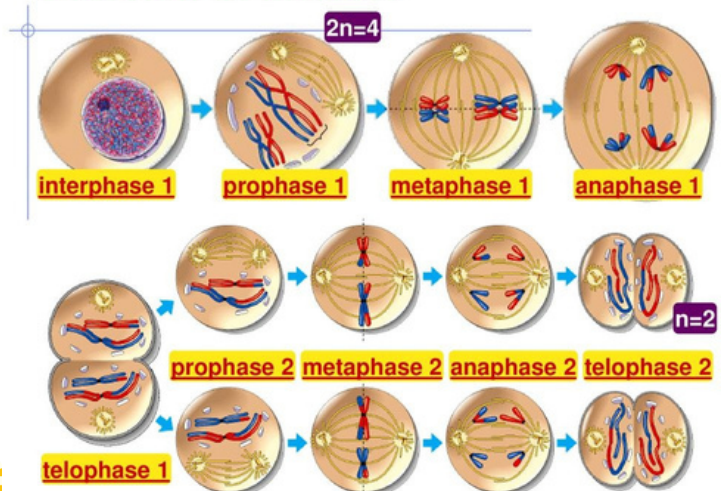
Animal cell
 Starts with appearance of furrow in the plasma membrane

Cell Cycle and Cell Division

Meiosis- Reductional Division

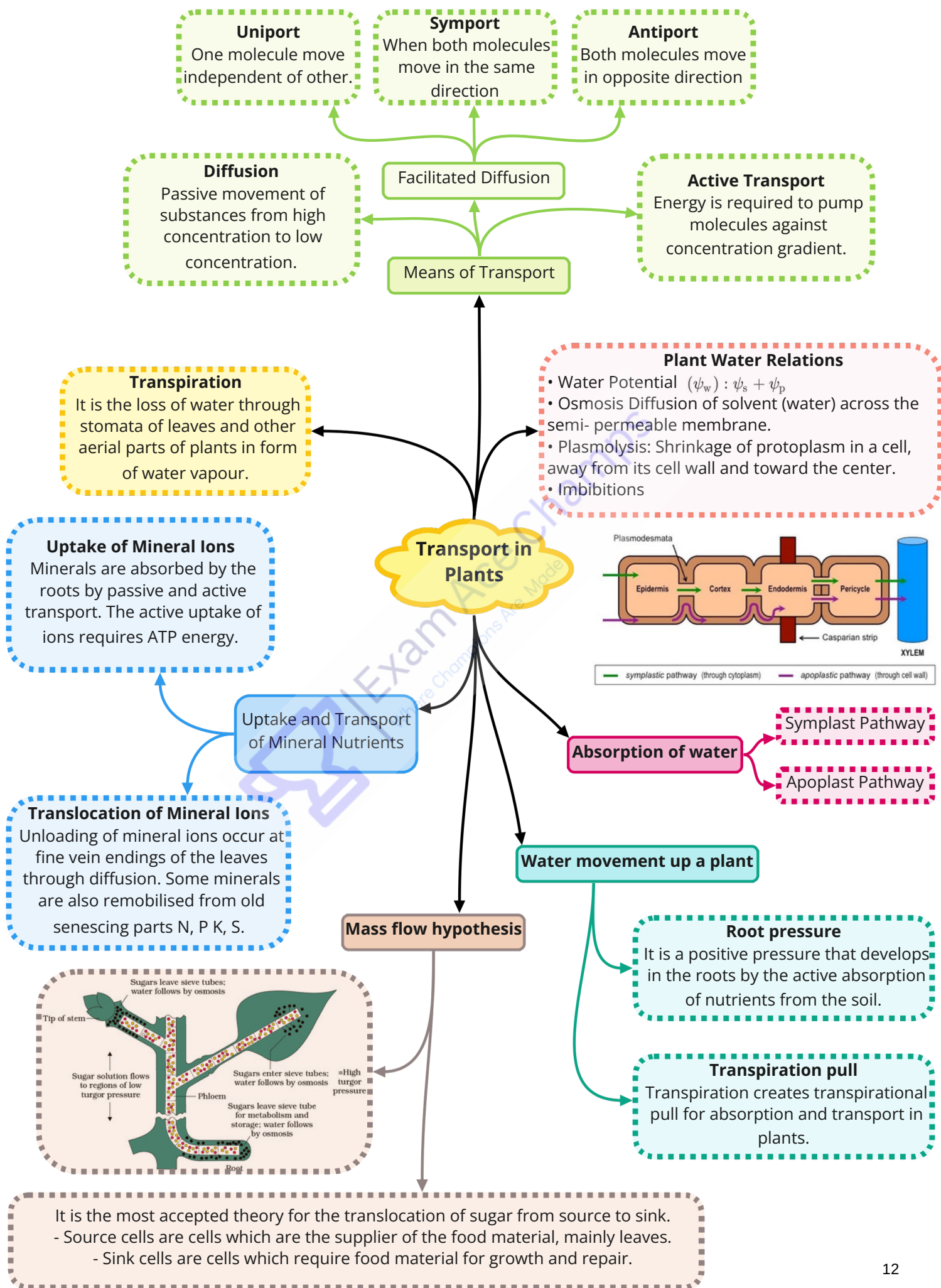


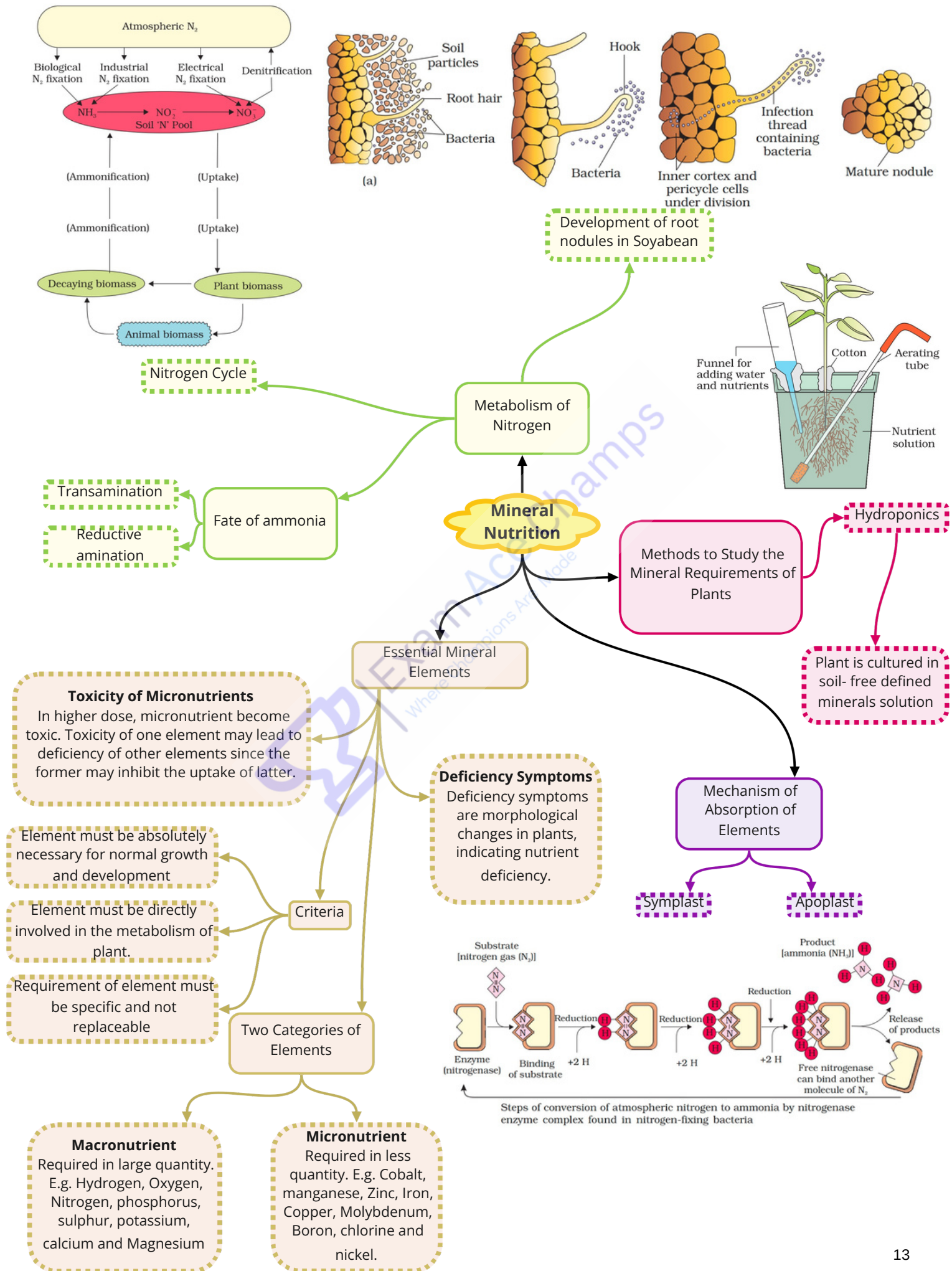
Overview of meiosis

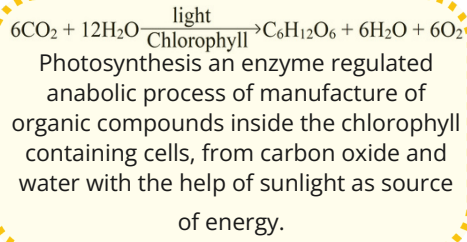
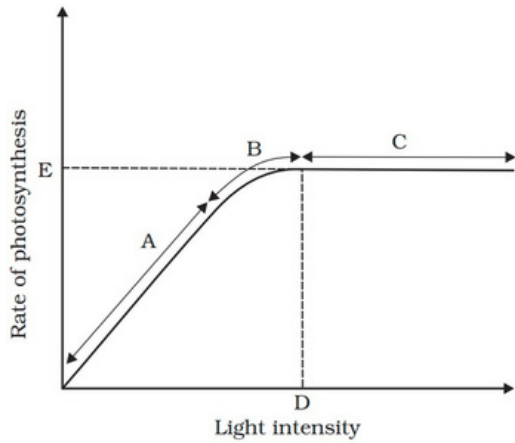


1. Leptotene (Duplicated chromosomes start to condense)
2. Zygotene (Synapsis begins)
3. Pachytene (Synapsis complete, Crossing over occurs)
4. Diplotene (Synaptonemal complex starts disappearing, Bchiasma visible)
5. Diakinesis (Bivalent ready for metaphase)

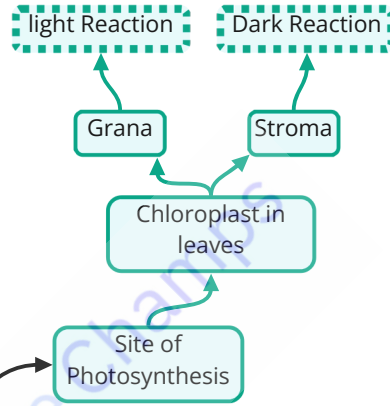
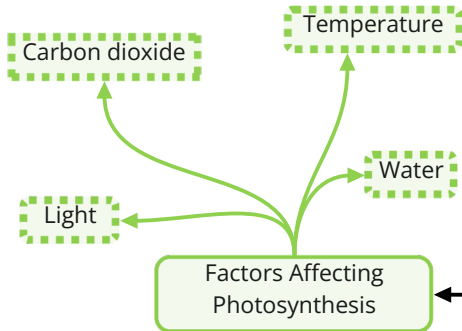
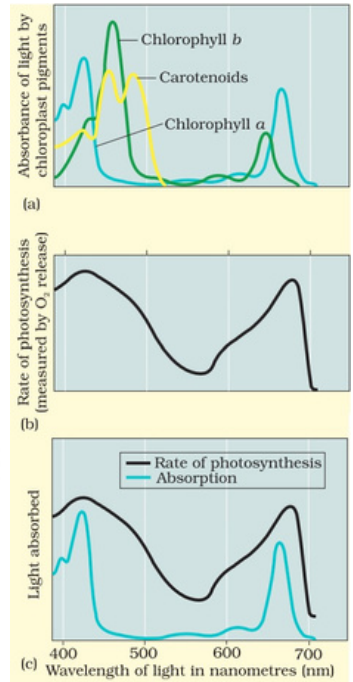
Class : 11th Biology
Chapter- 11 : Transport in Plants







Photosynthesis as a Means of Autotrophic Nutrition



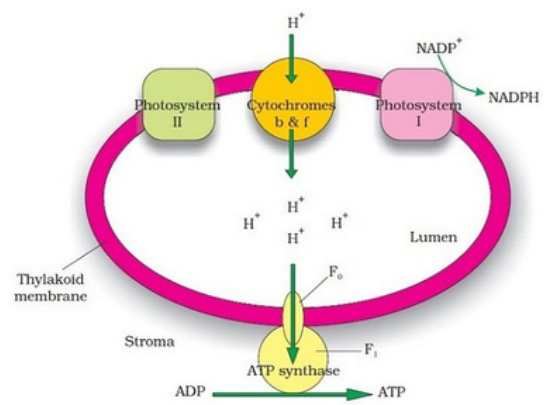
Photosynthesis in Higher Plants

- Pigment Involved in Photosynthesis**
- Chl a
 - Chl b
 - Xanthophyll
 - Carotenoids

Photorespiration

- It is the light induced respiration in green plants.
- It is a process in which there is no formation of ATP or NADPH, but there is utilization of ATP with release of CO_2

Chemiosmotic Hypothesis
 Explains the mechanism of ATP synthesis in thylakoid of chloroplast



Light Reaction

Cyclic

- Involves only photosystem I.
- Only ATP is formed.
- Electrons move in a closed circle.

Non- cyclic

- Involves both photosystem I and II.
- Both ATP and NADPH are produced.
- Electrons do not move in a closed circle.

Biosynthetic Phase Or Dark reaction

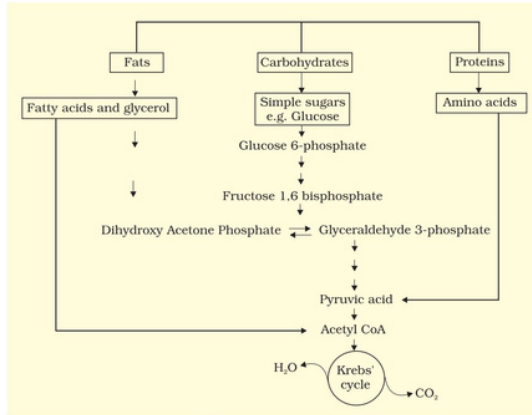
C4 or Hatch and Slack pathway

- Occurs in mesophyll and bundle sheath cells.
- Enzyme RuBisCo is absent from the mesophyll cells

C3 or Calvin cycle

- Carboxylation
- Reduction
- Regeneration
- For every CO_2 molecules, 3 molecules of ATP and 2 molecules of NADPH are required.

$$RQ = \frac{\text{Volume of CO}_2 \text{ Consumed}}{\text{Volume of CO}_2 \text{ Evolved}}$$



R.Q. values = 1 (for carbohydrate...
R.Q. = 0.7... R.Q. = 0.9...

R.Q formula

Respiratory Quotient

Fermentation
(Incomplete oxidation of pyruvic acid, under anaerobic condition)

Definition

Respiration is an energy releasing enzymatically controlled catabolic process which involves a step- wise oxidative breakdown of food substances inside living cells

Respiration in Plants

Amphibolic Pathway

Respiration is termed as amphibolic pathway as it involves both anabolism and catabolism

Aerobic respiration

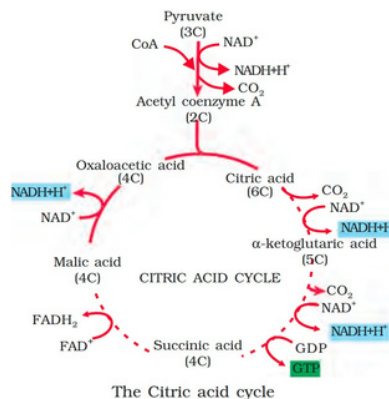
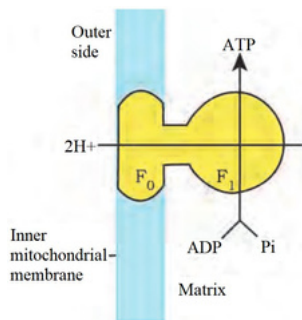
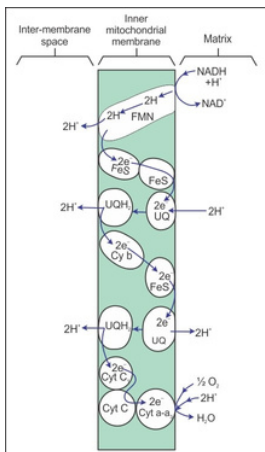
TCA Tricarboxylic Acid (Matrix of Mitochondria)

Glycolysis Or EMP pathway (Site: Cytoplasm)

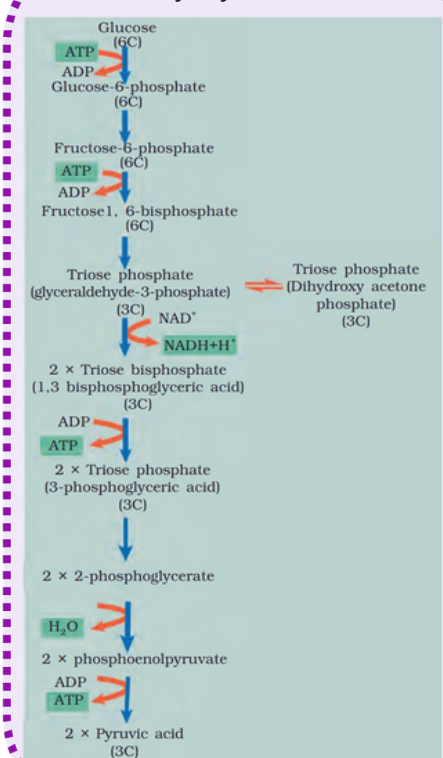
2nd of - Electron transport chain (Site: Inner mitochondrial membrane)

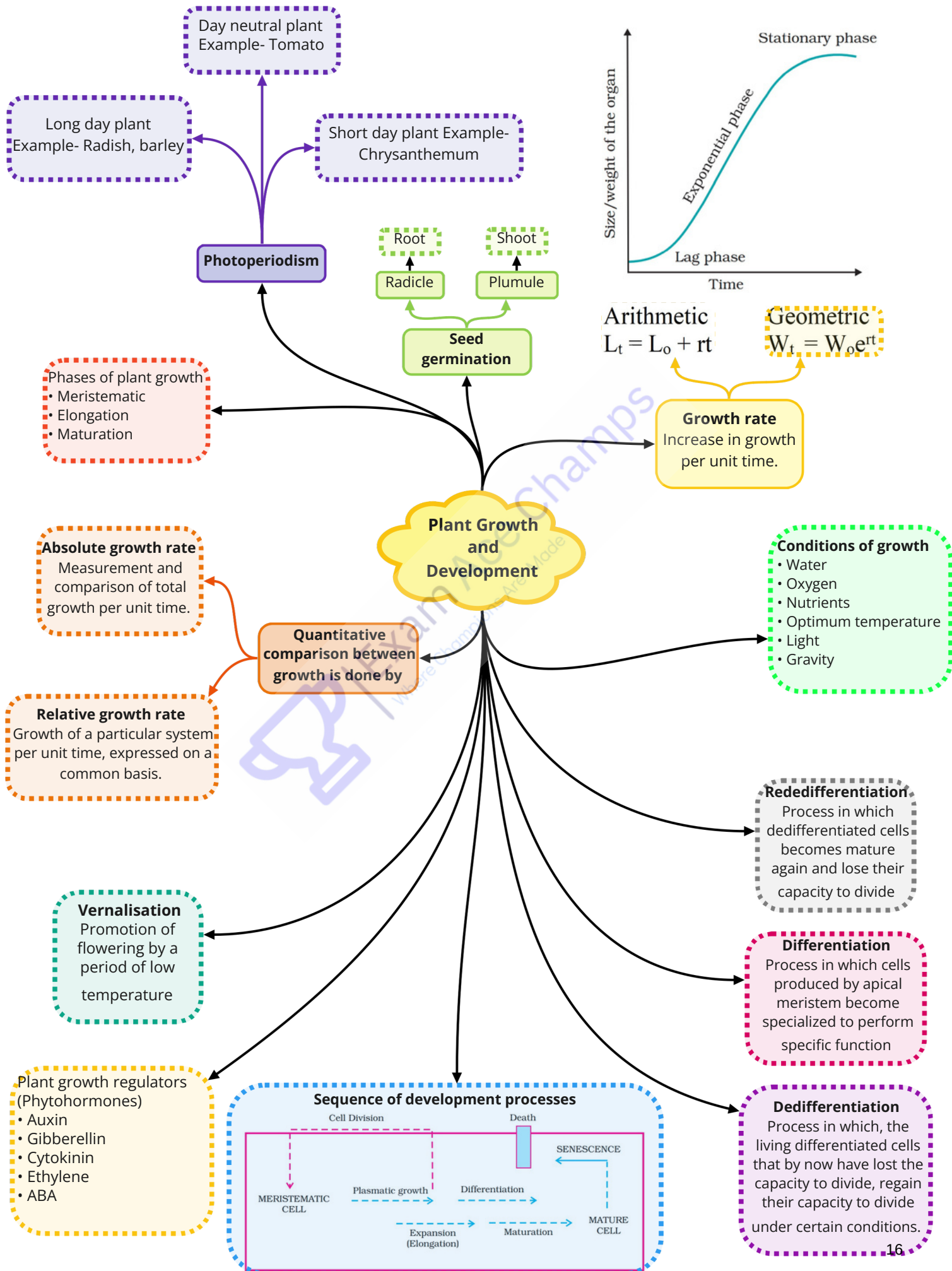
1st of - Oxidative Phosphorylation Site: F₀- F₁ particles in the inner mitochondrial membrane.

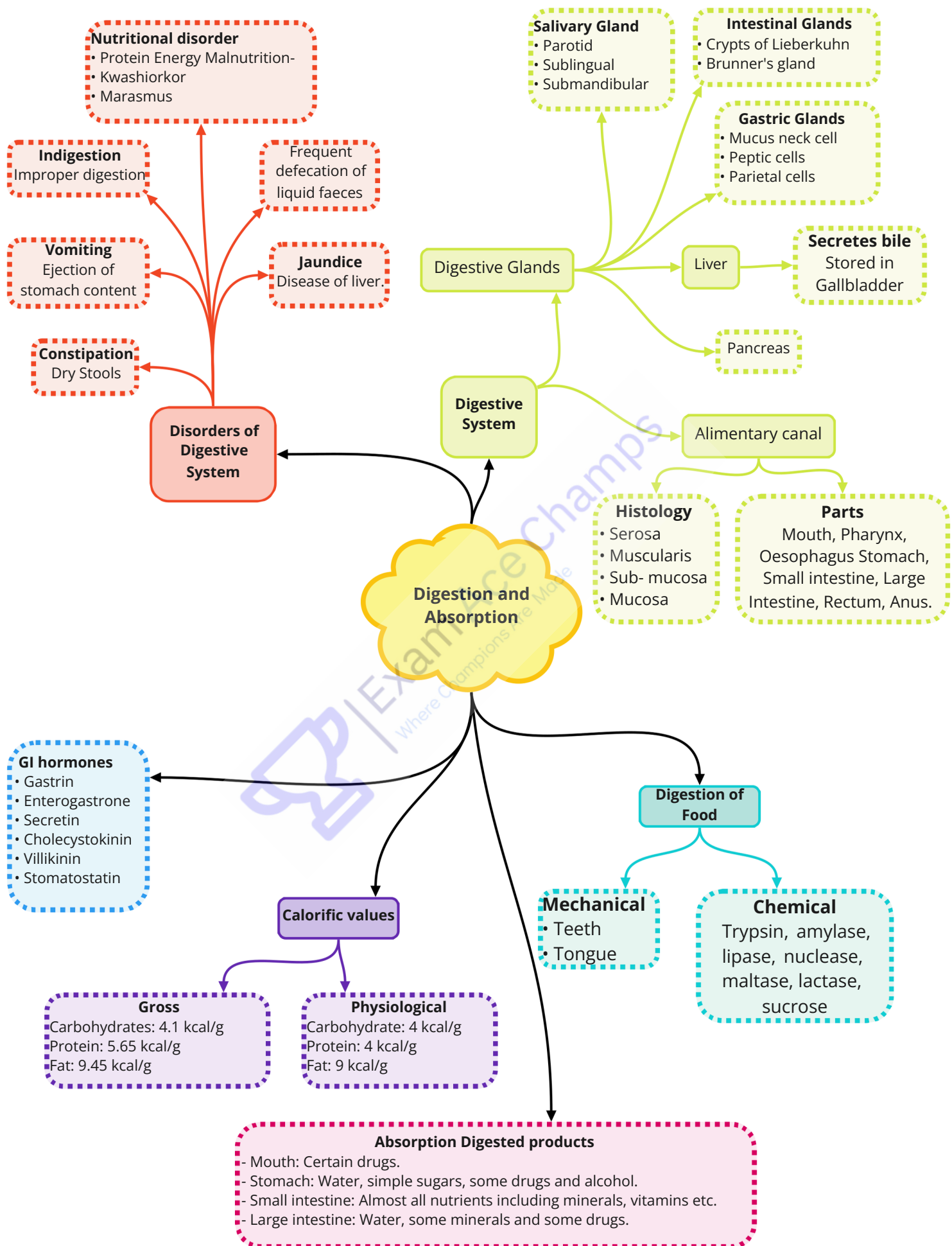
3rd of - TCA/ Tricarboxylic Acid (Site: Matrix of mitochondria)

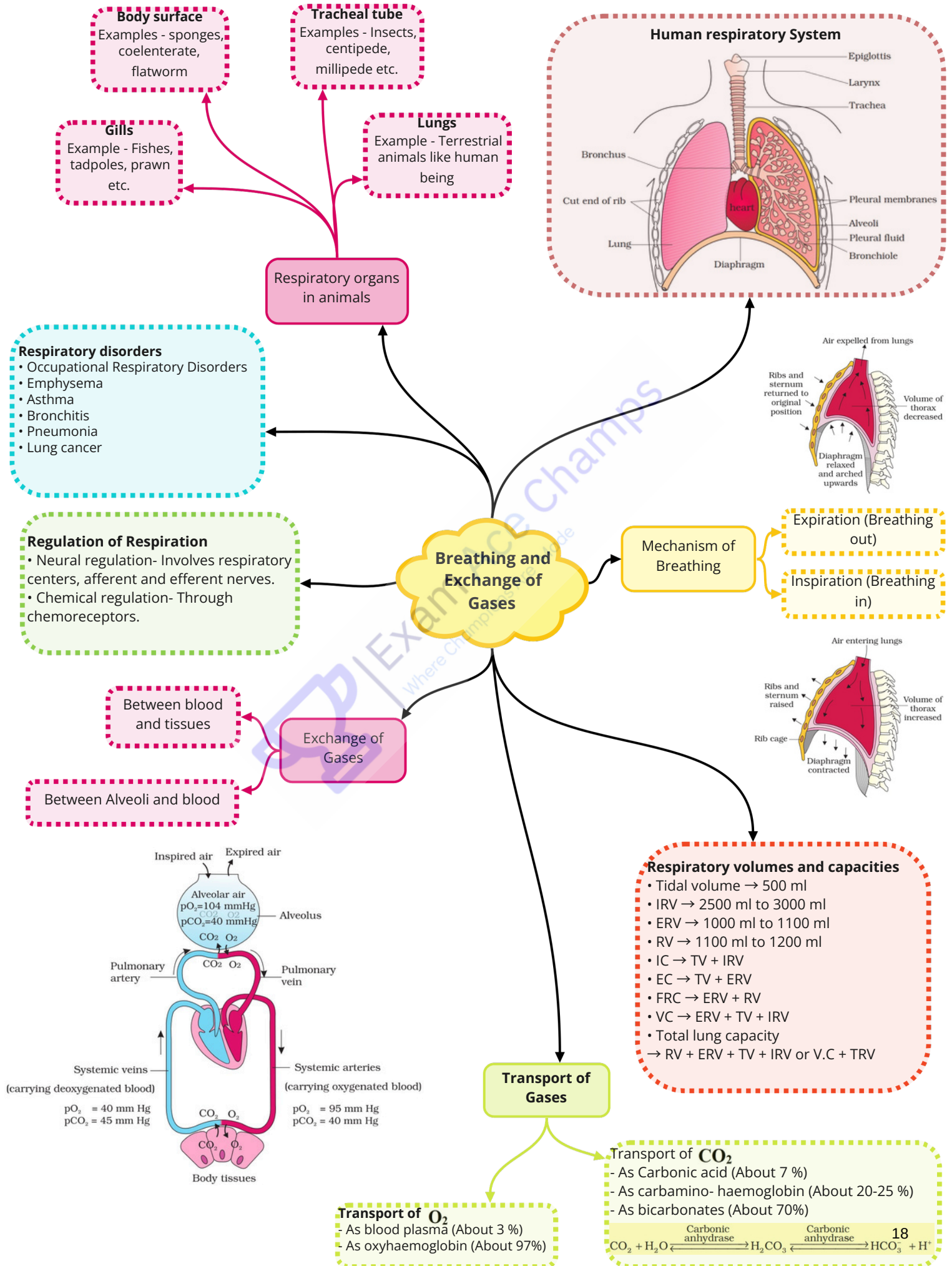


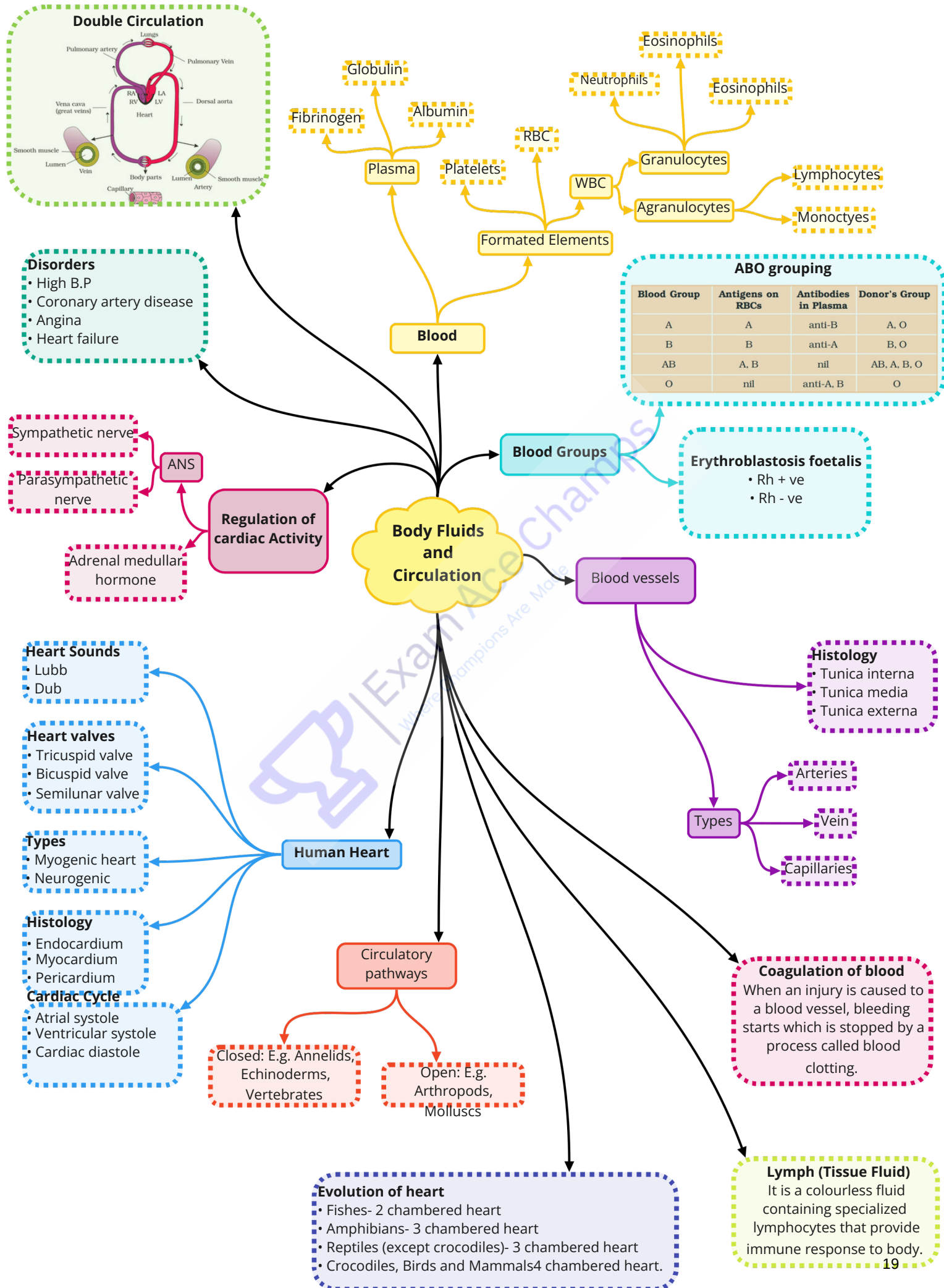
Glycolysis box











Ammonotelism

- Excretory product is ammonia
- Example- bony fishes, aquatic amphibians and insects

Ureotelism

- Excretory product is Urea
- Example- mammals, Terrestrial Amphibians

Uricotelism

- Excretory product is uric acid
- Example- Birds

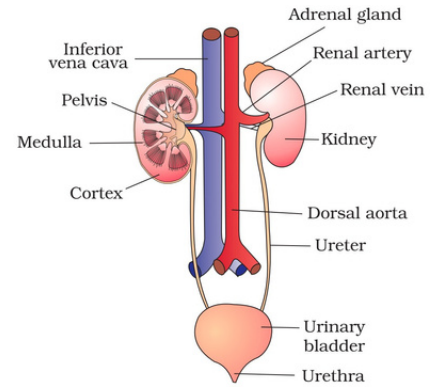
Disorders of the excretory System

- Uremia • Renal failure • Renal Calculi • Glomerulonephritis

Micturition

The process of expulsion of urine from urinary bladder

Modes of Excretion



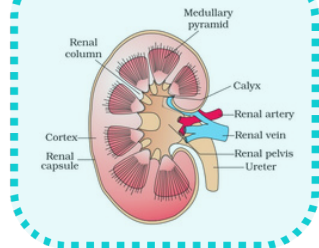
Urine Formation

- Glomerular filtration
- Tubular reabsorption
- Tubular secretion

Human Excretory System

- A Pair of Kidneys
- Ureter
- Urinary Bladder
- Urethra

Structure of kidney



Excretory Products and their Elimination

Function of the Tubules

- PCT 70% of electrolytes are reabsorbed
- Henle's Loop Maintains high osmolarity of medullary fluid
- DCT Conditional reabsorption of Na⁺ and water
- Collecting Duct Large amount of water is reabsorbed to produce concentrated urine

Regulation of kidney

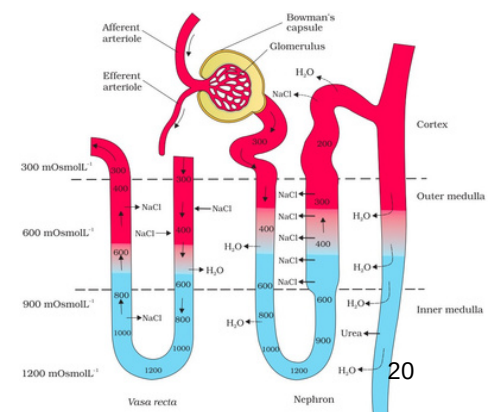
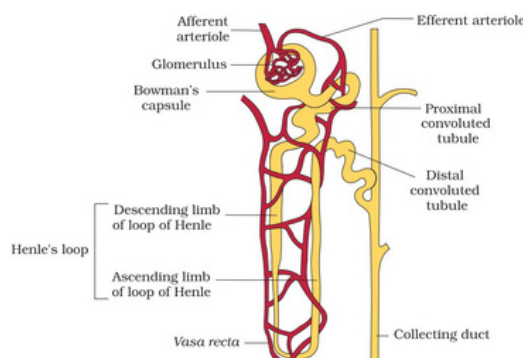
- By ADH (Vasopressin)
- By Juxtaglomerular Apparatus (Renin-Angiotensin mechanism)
- By ANF (in heart)

Other organs in excretion

- Liver
- Lungs
- Skin

Counter- current mechanism

- Henle's loop
- Vasa rectae



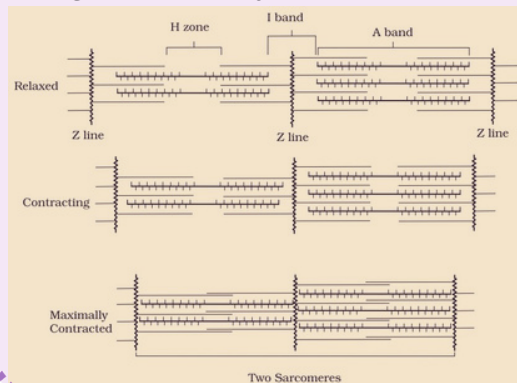
Properties of muscle fibers

- Excitability
- Contractility
- Extensibility
- Elasticity

Types of muscles

- Skeletal
- Visceral
- Cardiac

Sliding filament theory of Muscle Contraction



Types of Movement

- Amoeboid
- Ciliary
- Muscular

Disorders of muscular and skeletal System

- Myasthenia gravis
- Muscular dystrophy
- Tetany
- Arthritis
- Osteoporosis
- Gout

Joints

Cartilaginous
(Slightly movable)

Fibrous
(Immovable)

Synovial (movable)

- Ball and socket joint
- Hinge joint
- Pivot joint
- Gliding joint
- Saddle joint

Locomotion and Movement

Skeletal System

Axial (80 bones)

Appendicular (126)

Skull (40 bones)

Sternum (1)

Vertebral column

Ribs (12 pairs)

- Cervical (5)
- Thoracic (12)
- Lumbar (5)
- Sacral (1 fused)
- Coccygeal (1 fused)

Limb bones

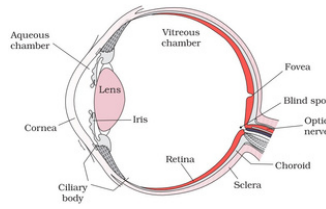
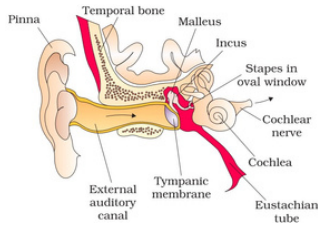
Girdles

Bones of hindlimbs
(30 x 2 = 60)

Bones of forelimbs
(30 x 2 = 60)

Pectoral girdle
(2 x 2 = 4)

Pelvic girdle (2)



Human Ear

- Outer Ear: Pinna, Ear canal, Tympanic membrane
- Middle Ear - Three ear ossicles
- Inner Ear - Utriculus, Sacculus and Cochlea

Human Eye

- Layers of eye -
- Outer layer - Sclera and Cornea
 - Middle layer- Choroid, Ciliary body, Iris, Pupil, lens
 - Innermost layer- Retina

Sense organs

Lower invertebrates

Neural organization is very simple

Insects

Brain is present along with number of ganglia and neural tissues

Vertebrates

More developed neural system

Neural System

Afferent Fibres

Efferent Fibres

Nerve Fibers

PNS

Parasympathetic

Sympathetic

Autonomic

Cranial nerves (12 pairs)

Spinal nerves (31 pairs)

Somatic

Human Neural System

CNS

Brain

Spinal cord

Neural Control and Coordination

Fore brain

- Cerebrum
- Thalamus
- Hypothalamus

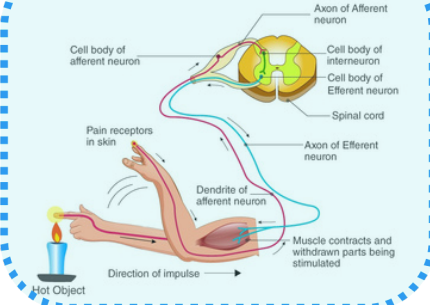
Mid brain

Hind brain

- Pons
- Cerebellum
- Medulla

Brain

Reflex action and Reflex arc



Generation and conduction of nerve impulse

- Maintenance of resting potential
- Action potential
- Propagation of action potential

Neuron (Structural and functional unit of neural system)]

Types of Neuron

Multipolar

Bipolar

Unipolar

Structure

Dendrite chain

Axon chain

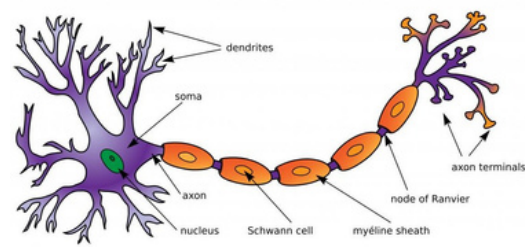
Cell body chain

Cytoplasm + typical cell organelles + Nissl's granules

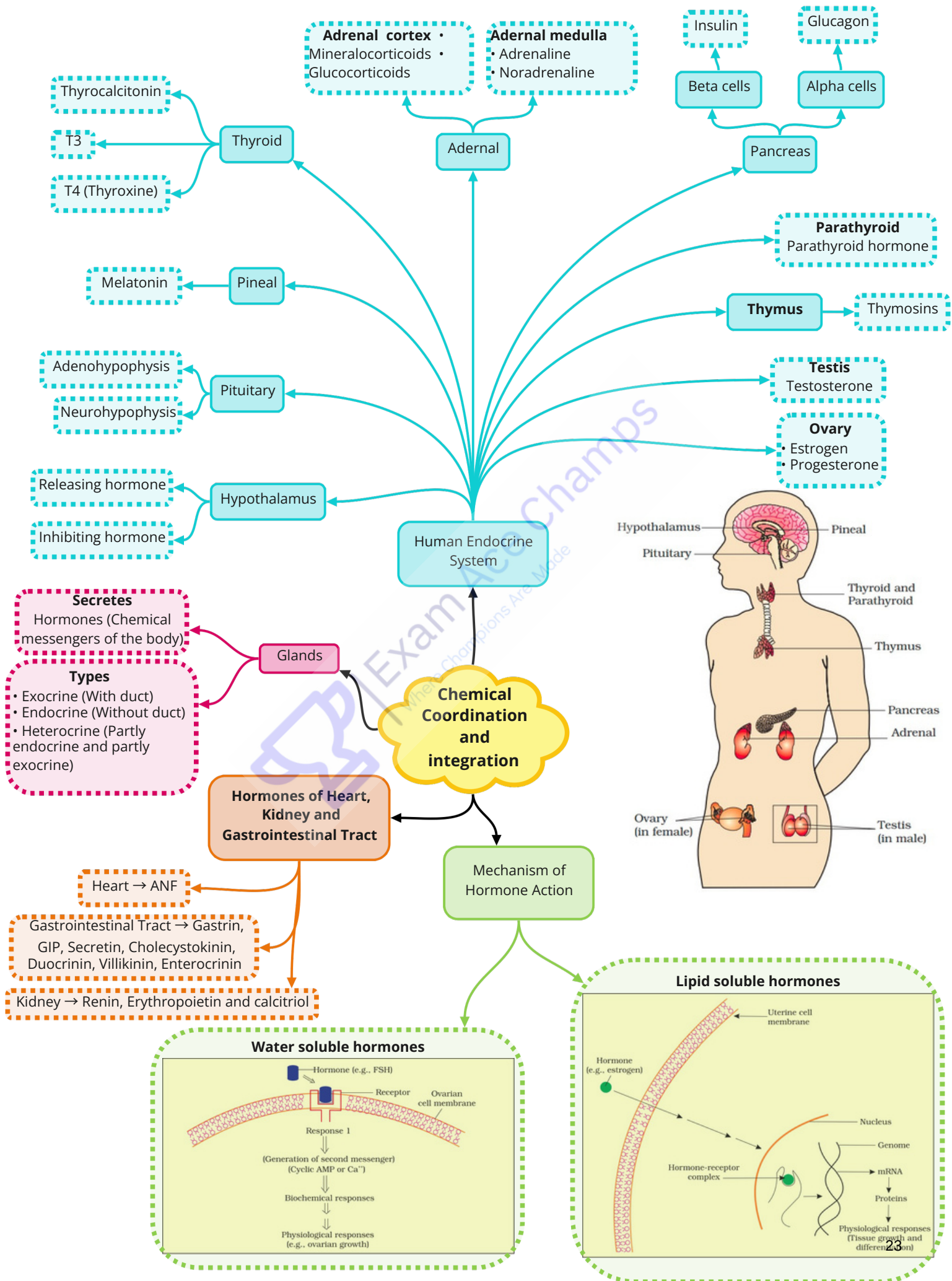
Myelinated axon

Non- myelinated axon

NEURON



Class : 11th Biology
Chapter- 22 : Chemical Coordination and integration





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